Total RNA Purification using Quick-RNA™ MagBead from Zymo Research

1. Add 200 µl (1 volume) RNA Lysis Buffer to 200 µl sample and mix well

2. Add 400 µl ethanol (95-100%) to the sample and mix well.

3. Add 30 µl MagBinding Beads and mix well for 20 minutes.

4. Transfer the plate/tube to the magnetic stand until beads have pelleted, then aspirate and discard the cleared supernatant.

5. Add 500 µl MagBead DNA/RNA Wash 1 and mix well the beads and discard the supernatant.

6. Add 500 µl MagBead DNA/RNA Wash 2 and mix well. Pellet the beads and discard the supernatant.

7. Add 500 µl ethanol (95-100%) and mix well. Pellet the beads and discard the supernatant.

8. Repeat step 7.

9. DNase I treatment (optional)

(D1) Add 50 µl DNase I Reaction Mix and mix gently for 10 minutes.

(D2) Add 500 µl RNA Prep Buffer and mix well for 10 minutes. Pellet the beads and discard the supernatant.

(D3) Repeat steps 7-8.

10. Dry the beads for 10 minutes or until dry.

11. To elute RNA from the beads, add ≥ 50 µl DNase/RNase-Free Water and mix well for 5 minutes.

12. Transfer the plate/tube to the magnetic stand until beads have pelleted, then aspirate and dispense the eluted RNA to a new plate/tube.

Reagent	Location
RNA Lysis Buffer	Reservoir 1 A1
Trash	Reservoir 3 A1
Ethanol	Reservoir 2 A1
Magnetic Beads	Reservoir 1 A2
DNA/RNA Wash 1	Reservoir 1 A3, Reservoir 1 A4
DNA/RNA Wash 2	Reservoir 1 A5, Reservoir 1 A6
DNase I mix	Reservoir 1 A7
RNA prep buffer	Reservoir 1 A8, Reservoir 1 A9
DNase/RNase-Free Water	Reservoir 1 A10

###INPUT SAMPLE NUMBER AND FINAL ELUTION VOLUME BELOW###

Sample number=

Elution Volume=

###INPUT SAMPLE NUMBER AND FINAL ELUTION VOLUME ABOVE###

NOTE: ALL MIXING STEPS ARE PERFORMED AT 1.8x SPEED (180 µL/s) TO ENSURE PROPER MIXING

CALCULATE NUMBER OF FULL COLUMNS

CALCULATE POSITION OF PARTIALLY FILLED COLUMN

CALCULATE Y LOCATION NEEDED TO PICK UP CORRECT NUMBER OF TIPS WITH PIPETTE

BUFFER TRACKING DEFINED IN FUNCTIONS

Step 1: Add 200 µl RNA Lysis Buffer to 200 µl Sample and Mix Well

- **If full columns of samples exist:**
  - Pick up a full column of transfer tips from column 1 of the transfer tip box.
  - Transfer 200 µL lysis buffer from reservoir A1 to the of each well in the full column on the prep plate.
  - Blow out into destination wells.
  - Return transfer tips to column 1 of the transfer tip box.
  - For each full column:

        - Pick up a full column of sample tips from the corresponding location in the left half of tip box 1.
        - Mix 150 µL 3 times in each well.
        - Return sample tips to the corresponding location in tip box 1.
- **If a partial column of samples exists:**
    - Pick up a partial column of transfer tips from the allocated y-coordinate in column 1 of the transfer tip box.
    - Transfer 200 µL lysis buffer from reservoir A1 to the top of the wells of the partially filled column on the prep plate.
    - Blow out into destination wells.
    - Return transfer tips to column 1 of the transfer tip box.
    - Pick up a partial column of sample tips from the corresponding location in the left half of tip box 1.
    - Mix 150 µL 3 times in each well.
    - Return sample tips to the corresponding location in tip box 1.

Step 2: Add 400 µL Ethanol to Each Sample

- **If full columns of samples exist:**
    - Pick up a full column of transfer tips from column 2 of the transfer tip box.
    - For each full column:
        - Repeat twice to add 400 µL total:
            1. Transfer 200 µL ethanol from reservoir A1 (with an air gap of 50 µL) to the top of each well in the full column.
            2. Blow out into destination wells.
    - Return transfer tips to column 2 of the transfer tip box.
    - For each full column:
        - Pick up a full column of sample tips from the corresponding location in the left half of tip box 1.
        - Mix 300 µL 5 times.
        - Return sample tips to the corresponding location in tip box 1.
- **If a partial column of samples exists:**

- Pick up a partial column of transfer tips from the allocated y-coordinate in column 2 of the transfer tip box.
- Repeat twice:
    - Transfer 200 µL ethanol from reservoir A1 (with an air gap of 50 µL) to the top of each well in the partial column.
    - Blow out into destination wells.
- Return transfer tips to column 2 of the transfer tip box.
- Pick up a partial column of sample tips from the corresponding location in the left half of tip box 1.
- Mix 300 µL 5 times in partial column of samples exists.
- Return sample tips to the corresponding location in tip box 1.

Step 3: Add 30 µL Magnetic Beads and Mix for 20 Minutes

- **If full columns of samples exist:**
    - Pick up a full column of transfer tips from column 3 of the transfer tip box.
    - Mix 30µL*10 then transfer 30 µL magnetic beads from reservoir A2 to the top of each well in the full column.
    - Blow out into destination wells.
    - Return the transfer tips to column 3 of the transfer tip box.
    - For each full column:
        - Pick up a full column of sample tips from the corresponding location in the left half of tip box 1.
        - Mix 200 µL 3 times in each well.
        - Return sample tips to the corresponding location in tip box 1.
- **If a partial column of samples exists:**
    - Pick up a partial column of transfer tips from the allocated y-coordinate in column 3 of the transfer tip box.
    - Mix 30µL*10 then transfer 30 µL magnetic beads from reservoir A2 to the top of the wells of the partially filled column.
    - Blow out into the destination wells.
    - Return the transfer tips to column 3 of the transfer tip box.

- Pick up a partial column of sample tips from the corresponding location in the left half of tip box 1.
- Mix partially filled column, 200 µL 3 times.
- Return sample tips to the corresponding location in tip box 1.

- **Loop for 20 minutes:**
  - **If full columns of samples exist:**
    - For each full column:
      - Pick up a full column of sample tips from the corresponding location in the left half of tip box 1.
      - Mix 200 µL 3 times in each well.
      - Return sample tips to the corresponding location in tip box 1.
  - **If a partial column of samples exists:**
    - Pick up a partial column of sample tips from the corresponding location in the left half of tip box 1.
    - Mix partially filled column, 200 µL 3 times.

Step 4: Transfer Samples to Magnetic Stand and Remove Supernatant

- **If full columns of samples exist:**
  - For each full column:
    - Pick up a full column of sample tips from the corresponding location in the left half of tip box 1.
    - Loop three times to transfer 830 µL:
      - Mix 150µL*3 then transfer 277 µL from the prep plate to the corresponding wells on the magnetic plate, with a 23 µL air gap.
      - Blow out into destination wells.
    - Return sample tips to the corresponding location in tip box 1.
- **If a partial column of samples exists:**
  - Pick up a partial column of sample tips from the corresponding location in the left half of tip box 1.
  - Loop three times to transfer 830 µL:
    - Mix 150µL*3 then transfer 277 µL from the prep plate to the corresponding wells on the magnetic plate, with a 23 µL air gap.

        - Blow out into destination wells.
    - Return sample tips to the corresponding location in tip box 1.
- **Pellet the magnetic beads:**
    - Engage the magnetic module at a height of 5.5.
    - Wait for 5 minutes to allow the beads to pellet.
- **Remove supernatant:**
    - **If full columns of samples exist:**
        - Pick up a full column of sample tips from the corresponding location in the left half of tip box 1.
        - Loop three times to transfer 798 µL, leaving 32 µL with the beads:
            - Transfer 266 µL from the prep plate to the corresponding wells on the magnetic plate, with a 30 µL air gap.
            - Blow out into destination wells.
        - Discard sample tips.
    - **If a partial column of samples exists:**
        - Pick up a partial column of sample tips from the corresponding location in the left half of tip box 1.
        - Loop three times to transfer 798 µL, leaving 32 µL with the beads:
            - Transfer 266 µL from the prep plate to the corresponding wells on the magnetic plate, with a 30 µL air gap.
            - Blow out into destination wells
        - Discard sample tips.

Step 5: Add 500 µL MagBead DNA/RNA Wash 1 and Discard Supernatant

- **If full columns of samples exist:**
    - Pick up a full column of transfer tips from column 4 of the transfer tip box.
    - For each full column:
        - Repeat twice to transfer 500 µL:
            - Check reservoir A3 has enough liquid – if not switch to A4

            - Transfer 250 µL of Wash 1 from reservoir-1 A3/A4 to the full column.
            - Blow out into destination wells.
    - Return transfer tips to column 4 of the transfer tip box.
    - For each full column:
        - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix 250 µL 5 times in each well.
        - Return sample tips to the corresponding location in tip box 1.
- **If a partial column of samples exists:**
    - Pick up a partial column of transfer tips from the allocated y-coordinate in column 4 of the transfer tip box.
    - Repeat twice to transfer 500 µL:
        - Check reservoir A3 has enough liquid – if not switch to A4
        - Transfer 250 µL of Wash 1 from reservoir-1 A3/A4 to the partial column.
        - Blow out into destination wells.
    - Return the transfer tips to column 4 of the transfer tip box.
    - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
    - Mix 250 µL 5 times in each well.
    - Return sample tips to the corresponding location in tip box 1.
- **Pellet magnetic beads:**
    - Engage the magnetic module at a height of 5.5 mm.
    - Wait for 5 minutes.
- **Remove supernatant:**
    - **If full columns of samples exist:**
        - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
        - Loop twice to transfer 500 µL:
            - Transfer 250 µL from full column to top of trash.
            - Blow out into destination wells.

        - Return sample tips to the corresponding location in tip box 1.
    - **If a partial column of samples exists:**
        - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
        - Loop twice times to transfer 500 µL:
            - Transfer 250 µL from full column to top of trash.
            - Blow out into destination wells
        - Return sample tips to the corresponding location in tip box 1.

Step 6: Add 500 µL MagBead DNA/RNA Wash 2 and Discard Supernatant

- **If full columns of samples exist:**
    - Pick up a full column of transfer tips from column 5 of the transfer tip box.
    - For each full column:
        - Repeat twice to transfer 500 µL:
            - Check reservoir A5 has enough liquid – if not switch to A6
            - Transfer 250 µL of Wash 2 from reservoir-1 A5/A6 to each well in the full column.
            - Blow out into destination wells.
    - Return transfer tips to column 5 of the transfer tip box.
    - For each full column:
        - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix 250 µL 5 times in each well.
        - Return sample tips to the corresponding location in tip box 1.
- **If a partial column of samples exists:**
    - Pick up a partial column of transfer tips from the allocated y-coordinate in column 5 of the transfer tip box.
    - Repeat twice to transfer 500 µL:
        - Check reservoir A5 has enough liquid – if not switch to A6
        - Transfer 250 µL of Wash 2 from reservoir-1 A5/A6 to the partial column.

            - Blow out into destination wells.
        - Return the transfer tips to column 5 of the transfer tip box.
        - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix 250 µL 5 times in each well.
        - Return sample tips to the corresponding location in tip box 1.
- **Pellet magnetic beads:**
    - Engage the magnetic module at a height of 5.5 mm.
    - Wait for 5 minutes.
- **Remove supernatant:**
    - **If full columns of samples exist:**
        - For each full column:
            - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
            - Loop twice to transfer 500 µL:
                - Transfer 250 µL from full column to trash.
                - Blow out into destination wells.
            - Return sample tips to the corresponding location in tip box 1.
    - **If a partial column of samples exists:**
        - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
        - Loop twice times to transfer 500 µL:
            - Transfer 250 µL from the partial column to the trash.
            - Blow out into destination wells
        - Return sample tips to the corresponding location in tip box 1.

Step 7 & 8: Perform Ethanol Wash (LOOPED)

Repeat twice:

- **If full columns of samples exist:**
    - Pick up a full column of transfer tips from column 2 of the transfer tip box.

- For each full column:
    - Repeat twice:
        - Transfer 250 µL ethanol from reservoir-2 A1 (with an air gap of 50 µL) to the top of the wells of the full column.
        - Blow out into destination wells.
    - Return the transfer tips to column 2 of the transfer tip box.
- Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
- Mix 250 µL 5 times in each well.
- Return sample tips to the corresponding location in tip box 1.

➢ **If a partial column of samples exists:**

- Pick up a partial column of transfer tips from the allocated y-coordinate in column 2 of the transfer tip box.
- Repeat twice:
    - Transfer 250 µL ethanol from reservoir-2 A1 (with an air gap of 50 µL) to the top of the wells of the partial column.
    - Blow out into destination wells.
- Return the transfer tips to column 2 of the transfer tip box.
- Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
- Mix 250 µL 5 times in each well.
- Return sample tips to the corresponding location in tip box 1.

➢ **Pellet magnetic beads:**

- Engage the magnetic module at a height of 5.5.
- Wait for 5 minutes.

➢ **Remove supernatant:**

- **If full columns of samples exist:**
    - For each full column:
        - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
        - Loop twice to transfer 500 µL:
            - ❖ Transfer 250 µL from the full column to trash.

            - Blow out after each transfer into the trash.
        - Return sample tips to the corresponding location in tip box 1.
    - **If a partial column of samples exists:**
        - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
        - Loop twice times to transfer 500 µL:
            - Transfer 250 µL from the partial column to trash.
            - Blow out after each transfer into the trash.
        - Return sample tips to the corresponding location in tip box 1.

Step 9: Add DNase Reaction Mix, mix for 10 minutes, add RNA Prep Buffer, mix for 10 minutes

- **Add DNase reaction mix:**
    - **If full columns of samples exist:**
        - Pick up a full column of transfer tips from column 6 of the transfer tip box.
        - For each full column:
            - Transfer 50 µL DNase reaction mix from reservoir A7 to the top of wells of the full column.
            - Blow out into the destination wells.
        - Return the transfer tips to column 6 of the transfer tip box.
    - **If a partial column of samples exists:**
        - Pick up a partial column of transfer tips from the allocated y-coordinate in column 6 of the transfer tip box.
        - Transfer 50 µL DNase reaction mix from reservoir A7 to the top of the wells in the partial column.
        - Blow out into destination wells.
        - Return the transfer tips to column 6 of the transfer tip box.
- **Loop for 10 minutes:**
    - **If full columns of samples exist:**
        - For each full column:

            - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
            - Mix 200 µL 3 times in each well.
            - Return sample tips to the corresponding location in tip box 1.
    - **If a partial column of samples exists:**
        - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix partially filled column, 200 µL 3 times.
- **Add RNA prep buffer:**
    - **If full columns of samples exist:**
        - Pick up a full column of transfer tips from column 7 of the transfer tip box.
        - For each full column:
            - Repeat twice to transfer 500 µL:
                - Check reservoir A8 has enough liquid – if not switch to A9
                - Transfer 250 µL of Wash RNA prep buffer from reservoir-1 A8/A9 to each well in the full column.
                - Blow out into destination wells.
        - Return the transfer tips to column 7 of the transfer tip box.
    - **If a partial column of samples exists:**
        - Pick up a partial column of transfer tips from the allocated y-coordinate in column 8 of the transfer tip box.
        - Repeat twice to transfer 500 µL:
            - Check reservoir A8 has enough liquid – if not switch to A9
            - Transfer 250 µL of Wash RNA prep buffer from reservoir-1 A8/A9 to each well in the partial column.
            - Blow out into destination wells.
        - Return the transfer tips to column 7 of the transfer tip box.
- **Loop for 10 minutes:**
    - **If full columns of samples exist:**
        - For each full column:

        - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix 200 µL 3 times in each well.
        - Return sample tips to the corresponding location in tip box 1.
    - **If a partial column of samples exists:**
        - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix partially filled column, 200 µL 3 times.

Repeat Step 7 & 8

Refer to step 7 & 8 – it is identical.

Step 11: Add DNase/RNase free water and Mix

- **If full columns of samples exist:**
    - Pick up a full column of transfer tips from column 8 of the transfer tip box.
    - For each full column:
        - Transfer Elution Volume µL DNase/RNase free water from reservoir A10 to the top of each well in the full columns on the magnetic plate.
        - Blow out into destination wells.
    - Return transfer tips to column 8 of the transfer tip box.
- **If a partial column of samples exists:**
    - Pick up a partial column of transfer tips from the allocated y-coordinate in column 8 of the transfer tip box.
    - Transfer Elution Volume µL DNase/RNase free water from reservoir A10 to the top of each well of the partial column on the magnetic plate.
    - Blow out into destination wells.
    - Return transfer tips to column 8 of the transfer tip box.
- **Loop for 5 minutes:**
    - **If full columns of samples exist:**
        - For each full column:

        - Pick up a full column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix full column, Elution Volume µL 3 times.
        - Return sample tips to the corresponding location in tip box 1.
    - **If a partial column of samples exists:**
        - Pick up a partial column of sample tips from the corresponding location in the right half of tip box 1.
        - Mix partially filled column, Elution Volume µL 3 times.

Step 12: Transfer Eluted RNA to Elution Plate

- **If full columns of samples exist:**
    - For each full column:
        - Pick up a full column of fresh tips from tip box 2.
        - Transfer Elution Volume µL eluted RNA from column magnetic plate to the corresponding wells on the elution plate.
        - Blow out into destination wells.
        - Discard the tips in the waste container.
- **If a partial column of samples exists:**
    - Pick up a partial column of fresh tips from the corresponding location in tip box 2.
    - Transfer Elution Volume µL eluted RNA from partial column of the magnetic plate to the corresponding column on the elution plate.
    - Blow out into destination wells.
    - Discard the tips in the waste container.
- **Disengage the magnetic module.**